

Cement manufacturing using alternative fuels and the advantages of process modelling

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Abstract

Energy costs and environmental standards encouraged cement manufacturers world-wide to evaluate to what extent conventional fuels can be replaced by alternative fuels, i.e., processed waste materials. Clinker burning is well suited for various alternative fuels. In order select a suitable alternative fuel, a commercial modelling tool (ASPEN PLUS®) is used to model the four-stage preheater kiln system of a full-scale cement plant (clinker production ~ 2900 tons/day), using petcoke as fuel. The goal is to optimise process control and alternative fuel consumption, while maintaining clinker product quality. Calculations with varying amounts of different fuels are compared with a reference case. The dependence of process performance on the amount of combustion air is clearly demonstrated and the energy demand of the process could be predicted for varying fuel mixes.

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1. Introduction

1.1. Cement plant process modelling

Recent studies on modelling cement manufacturing reported in the literature are mostly based on computational fluid dynamics (CFD), and they mainly study aerodynamic behaviour of particles in the preheating system, the shape and temperature of the flame in the combustion zone, coal combustion itself as well as oxygen enrichment in the

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burning zone and not so much the thermodynamics and clinker chemistry taking place in the kiln system [1–3]. This paper describes the simulation of a cement clinker production plant using the commercial software Aspen Plus®, with a strong focus on clinker chemistry and thermodynamics in the rotary kiln and the effect of alternative fuels on material flows, emissions and product quality.

1.2. The cement manufacturing process

Cement manufacturing consists of raw meal grinding, blending, pre-calcining, clinker burning and cement grinding. In short, limestone and other materials containing calcium, silicon, aluminium and iron oxides are crushed and milled into a raw meal. This raw meal is blended (in for instance blending silos) and is then heated in the pre-heating system to initiate the dissociation of carbonate to calcium oxide and carbon dioxide. A secondary fuel is fed into the preheating system to keep the temperature sufficiently high. The meal then proceeds to the kiln for heating and reaction between calcium oxide and other elements to form calcium silicates and aluminates at a temperature up to 1450 °C. Primary fuel is used to keep the temperature high enough in the burning zone for the chemical reactions to take place. The reaction products leave the kiln as a nodular material called clinker. The clinker will be inter-ground with gypsum, limestone and/or ashes to a fine product called cement [4].

Fig. 1 shows a cement manufacturing process from raw material quarrying to the bagging of the cement. The waste tyre particles are fed into the lower part of the kilns pre-heating system, hereafter referred to as the riser duct [5].

1.3. Kiln system chemistry

The chemical reactions that occur in the kiln are described in detail in Ref. [6]. The temperature is increased when going from the meal feed to the rotary kiln. The most

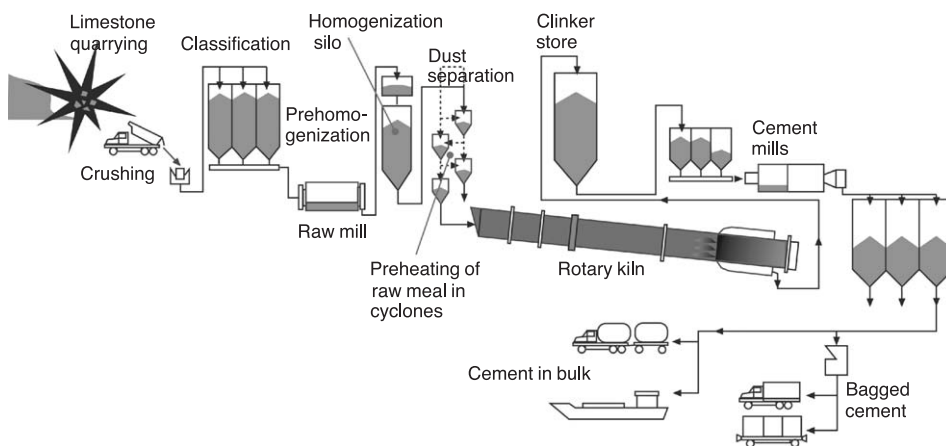


Fig. 1. Cement manufacturing process.

Table 1
Alternative fuel options for the cement industry

Liquid waste fuels	Tar, chemical wastes, distillation residues, waste solvents, used oils, wax suspensions, petrochemical waste, asphalt slurry, paint waste, oil sludge
Solid waste fuels	Petroleum coke (“petcoke”), paper waste, rubber residues, pulp sludge, used tires, battery cases, plastic residues, wood waste, domestic refuse, rice chaff, refuse derived fuel, nut shells, oil-bearing soils, sewage sludge
Gaseous waste	Landfill gas, pyrolysis gas

important oxides that participate in the reactions are CaO, SiO₂, Al₂O₃ and Fe₂O₃. First, water is removed from the meal. In the preheating section (700–900 °C) calcination as well as an initial combination of alumina, ferric oxide and silica with lime takes place. Between 900 and 1200 °C belite, C₂S (=2CaO*SiO₂), forms. Above 1250 °C, a liquid

Table 2
Properties of fuels of interest to the cement industry (partly from Ref. [8])

Dry, in wt.% (unless otherwise indicated)	Bituminous coal (Poland)	Petcoke	Meat and bone meal	Sewage sludge	Car tyre scrap <i>Nokia/Slite average</i>	Car tyre rubber	Coal–petcoke mix
C	70.6	89.5	42.1	40.5	79.6	87.0	75.1
H	4.3	3.08	5.83	7.0	7.0	7.82	4.20
N	1.2	1.71	7.52	0.84	0.43	0.33	1.70
S	1.3	4.00	0.38	0.12	1.54	0.80	3.00
O ^a	11.8	1.11	15.3	25.2	5.0	1.81	4.90
Cl	0.07		0.20	1.0	0.063		
P (P ₂ O ₅ in ash)	0.016		(41.2)		0.015		(0.046)
Na (Na ₂ O in ash)	0.05		(1.47)		0.04	(0.9)	(2.4)
K (K ₂ O in ash)	0.12		(0.28)		0.038	(1.1)	(2.8)
Ca (CaO in ash)	0.18		(18.4)	1.83	0.53	(8.7)	(11.7)
Fe (Fe ₂ O ₃ in ash)	0.31		(0.14)		0.36	(1.7)	(5.4)
Al (Al ₂ O ₃ in ash)	1.07		(0.02)		0.11	(16.6)	(1.2)
Si (SiO ₂ in ash)	2.0		(0.45)		0.94	(27.8)	(55.8)
Mg (MgO in ash)	0.08			0.32		(1.2)	(2.6)
Ni (NiO in ash)		0.0432			0.018	(0.0065)	(0.044)
Hg (mg/kg)		0.05			<0.1		
Pb (PbO in ash)		0.00053			0.004	(0.0373)	(0.0054)
Zn (ZnO in ash)		0.00103			1.48	(56.6)	(0.0092)
Mn (MnO in ash)						(0.021)	(0.027)
V (V ₂ O ₅ in ash)		0.0906				(0.0043)	(0.251)
As (mg/kg)					0.7		
Cr (Cr ₂ O ₃ in ash)		0.0013			(0.034)	(0.0008)	(0.0018)
Cd (mg/kg)		0.46			8		
Co (CoO in ash)		0.0001				(0.495)	(0.00017)
Ash	11.1	0.50	28.3	17.9	6.35	2.20	11.1
Volatiles	35.9	10.0	64.5	85.0	69.4	66.6	20.0
C-fix	53.0	89.5	7.20	5.00	30.6	31.1	69.2
H ₂ O (wt.%)	3.0	1.50	8.09	5.20	0.66	0.73	1.30
LHV (MJ/kg)	27.4	33.7	16.2	15.8	35.5	35.6	29.0
HHV (MJ/kg)	28.4					37.3	29.7

^a Measured, i.e., not by difference.

phase appears and this promotes the reaction between belite and free lime to form alite, C_3S ($=3CaO \cdot SiO_2$). During the cooling stage, the molten phase forms C_3A , tri-calcium aluminate ($=3CaO \cdot Al_2O_3$) and a ferrite phase, normally C_4AF ($=4CaO \cdot Al_2O_3 \cdot Fe_2O_3$). If the cooling is slow, alite may decompose and appear as secondary belite and free lime. Usually, the production of clinker is done so that one type of clinker allows the plant to manufacture several well-defined types of cement that comply with the physical demands as specified by cement standards.

1.4. Alternative fuels

The range of fuels is extremely wide. Traditional kiln fuels are gas, oil or coal. Materials like waste oils, plastics, auto shredded residues, waste tyres and sewage sludge (SS) are often proposed as alternative fuels for the cement industry. Also, all kinds of slaughterhouse residues are offered as fuel nowadays.

To be able to use any of these fuels in a cement factory, it is necessary to know the composition of the fuel. The choice is normally based on price and availability. The energy and ash contents are also important, as are the moisture and volatiles contents. All kinds of varieties from liquid to solids, powdered or as big lumps can be encountered when dealing with alternative fuels, requiring a flexible fuel feeding system. Somehow, they should all be fed into the burning chamber of the process. It may be fed directly into the burning zone in the kiln itself or into the pre-heating system for dissociating part of the carbonates from the meal before it enters the kiln for clinker formation. In Table 1, we can see examples of different alternative fuels. They are separated into three groups [4].

In Table 2, several fuels of interest to the cement industry and their properties are listed. Some of these were used in model calculations reported here. The calculations were made to test the influence of a fuel change on the kiln process, specially the demand of combustion air in the burning zone.

2. Process calculations

The process for which the air calculations reported in this paper are done (one of the cement plant of the CRH) consists of a rotary kiln and a cyclone string. The pre-heating system consists of four cyclones and a riser duct. The rotary kiln is producing almost 3000 tons clinker/day, and so far the main fuel has been petcoke. All these different parts in the process are modelled with different reactors that are defined in the simulation program. The process model (set up under Aspen Plus® 10.2) consists basically of mass balances for compounds and reactors, the flowsheet as used in the calculations is shown in Fig. 2. A lot of information on the design and operation of the cement plant is used in the modelling work [7]:

- Temperatures and pressures at various locations and of the incoming mass streams
- The dimensions and operational parameters of the cyclones and the electrostatic precipitators (ESPs), defining their grade efficiency performance

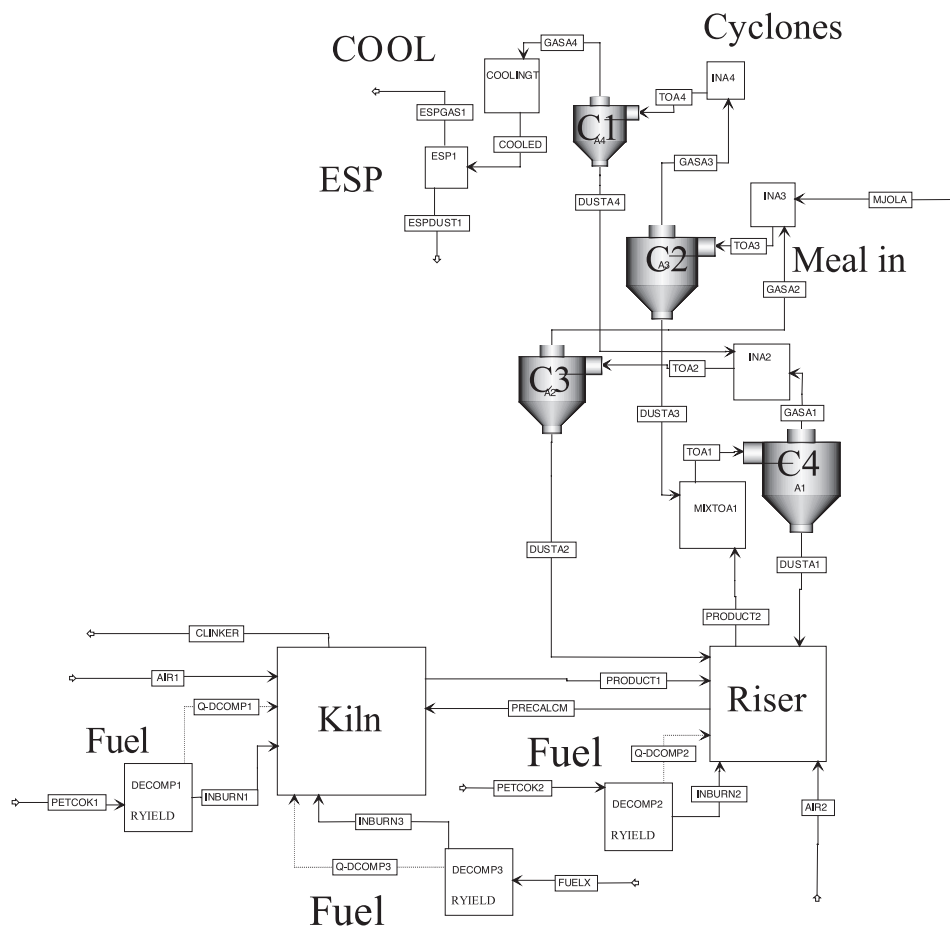


Fig. 2. Process simulation flowsheet.

- Chemical composition and heating values of the incoming raw meal, the primary and secondary fuel, with different particle size distributions for all materials
- Incoming mass flows of raw meal, primary and secondary fuel and combustion air
- Cooling equipment heat fluxes

The blocks are chosen such that the chemistry in the different parts of the process can be specified as realistic as possible (e.g., equilibrium or non-equilibrium reactors), in a user-friendly way. After having connected the various modelling blocks, representing different pieces of equipment in the clinker manufacturing process, the flow sheet of the model is rather similar to the flow sheet of the process.

The final target is to be able to describe the behaviour of the kiln process in a realistic way, when the fuel is fully or partly replaced by an alternative fuel. The outcome could possibly be called “quality control” giving vital information about the

chemical changes caused by the alternative fuel, the heat transmission in the kiln and possibly also other parameters controlling the clinker burning process. The material streams from one reactor or block to another, the in- and out-going flows are shown, as are also different (alternative) fuel feeding locations. The feed of alternative fuels is possible to both the rotary kilns burning zone and to the pre-heating system, in this case meaning the riser duct with the pre-calciner cyclones. Chemical analyses are needed both for the kiln feed and the fuels that are intended to be tested with this model.

The model is first run with the “normal operation” values (reference case) meaning that only petcoke is used as fuel both in the burning zone and in the riser duct (primary and secondary fuel). The pre-heated air to the burning zone is varied between 430 000 and 500 000 m³/h (800 °C, 1 bar). Calculations with alternative fuels are done by partly replacing the primary or secondary fuel. Meat and bone meal (MBM) and sewage sludge are considered: their amounts are varied, giving a base load of heat into the kiln system that is roughly that of the reference case. The raw meal feed is according to process values for both chemical composition and amounts.

3. Results

Fuels involved in the calculations are petcoke, MBM and SS. Their analyses are shown in Table 2. Depending on the cement plant's location, some alternative fuels may be more favourable than others; important issues are for instance transport costs and the availability of the fuel.

First, the reference case was calculated, i.e., 12 500 and 1500 kg/h petcoke as primary and secondary fuel, respectively. The combustion air requirement was calculated to a volume that gives 2% oxygen in the flue gases after the burning zone. This value is considered preferable with respect to CO emissions and ESP filter performance. For the reference case, Fig. 3 shows that the need of preheated air for combustion is close to 440 000 m³/h. All other calculated cases are evaluated against this reference case (only petcoke as fuel). Fig. 2 also shows that using 1500 kg/h MBM or SS as secondary fuel instead of petcoke, gives a lower demand of combustion air compared to the reference case (the level of 2% oxygen after the burning zone is reached with less air). The result is the opposite when the fuel feed of MBM or SS is 3000 kg/h. The primary fuel is petcoke in these cases; the combinations with the secondary fuels were varied as to get a total heat input to the kiln system that is higher or lower than for the reference case.

The situation is slightly different when alternative fuels (3000 kg/h MBM or SS) are fed directly into the burning zone and replacing part of the primary fuel petcoke. The secondary fuel is petcoke as well (1500 kg/h) in these calculations. The reference case involves 12 500 kg/h petcoke as primary fuel, which was lowered to 11 500 and 10 000 kg/h, respectively. These differences in petcoke and overall fuel input amounts explain the differences seen in Fig. 4.

After calculating the air amounts to the burning zone, the total heat input to the kiln system (kiln and riser duct) was calculated for different fuel combinations when

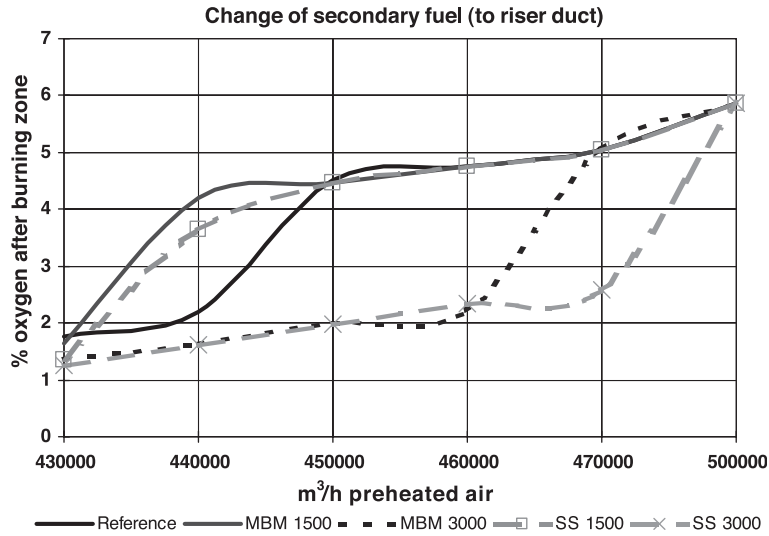


Fig. 3. Plotted results for different fuel combinations when the secondary fuel is partly replaced by an alternative fuel (primary fuel: petcoke).

compared with the reference case. These calculations were made for two daily clinker production values to validate the efficiency of the kiln system—see Fig. 5. Compared to the reference case, different fuel combinations may give a lower or higher energy input

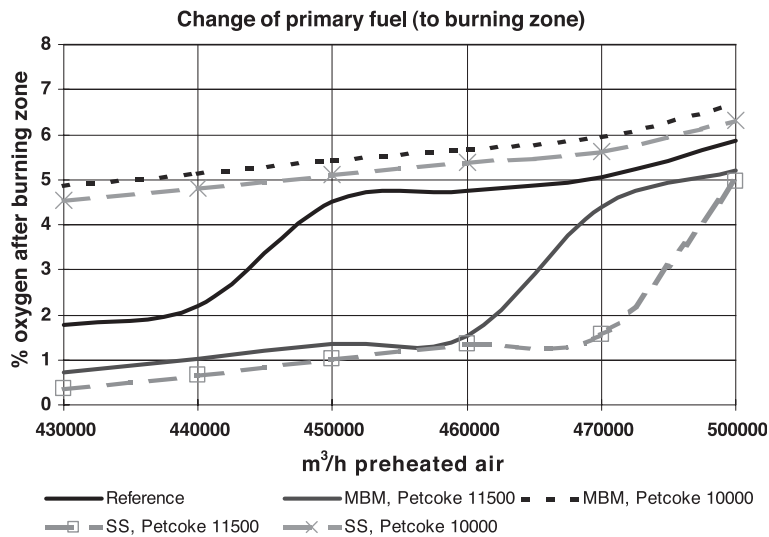


Fig. 4. Results for different fuel combinations when the primary fuel petcoke is partly replaced by an alternative fuel (alternative fuel 3000 kg/h, petcoke kg/h as indicated).

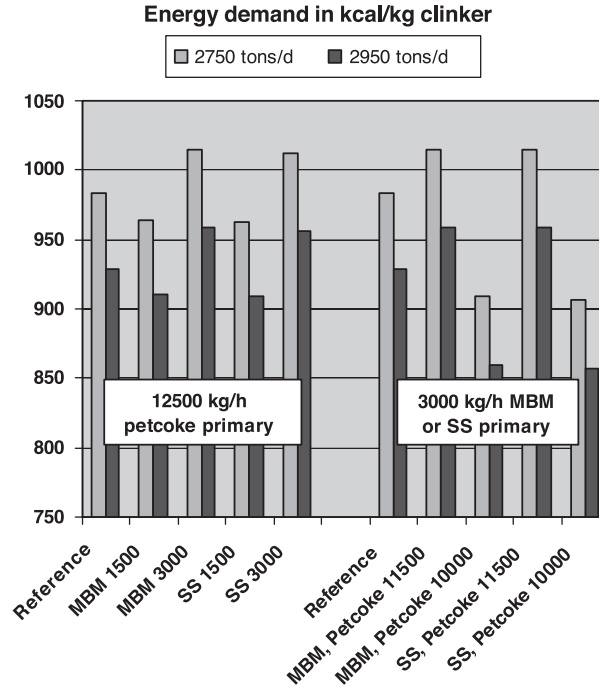


Fig. 5. Energy demand for different fuel combinations for two amounts of daily clinker production.

(demand). In Fig. 5, the energy consumption values appear somewhat high for the reference case and may be slightly incorrect.

4. Discussion and conclusions

When part of the primary or secondary fuel of a cement plant (here: petcoke) is to be replaced by alternative fuels, it must be considered beforehand whether the air feeding system (kiln exhaust fan) allows for that. For alternative fuels like MBM and SS, it is calculated here that a slightly higher need of air is to be allowed for when operating with larger amounts of alternative fuel fed to the riser. Although it is only a matter of around 3–4% higher need, problems may arise especially when setting the operational values for the process (fan speed, production levels, dust emissions). If there is fan capacity enough to take care of the 3–4% higher need of air, there will not be any difficulties when switching between conventional and alternative fuels. With 1500 kg/h MBM or SS as secondary fuel, the need of combustion air is about 2% less than in the reference case. The energy input is then only 20 kcal/kg lower than for the reference case. This case may allow for a slightly higher production level.

If alternative fuel is fed to the burning zone and replacing part of the primary fuel the results show that approximately 5–10% more air is needed for combustion for cases MBM

3000/Petcoke 11 500 kg/h and SS 3000/Petcoke 11 500 kg/h primary fuel. These are also quite close to the reference case when it comes to energy input to the kiln. Cases MBM 3000/Petcoke 10 000 kg/h and SS 3000/Petcoke 10 000 kg/h show a very positive result when it comes to air demand, but the energy value is very low when compared with the reference case. On the other hand, this would allow for a higher use of petcoke to supply the extra energy that is needed when increasing the production, and make use of the excess of air in the combustion zone.

When calculations show that the air demand is higher with the new fuel than for the reference case, it means that the kiln fans must run at higher speed to supply the larger air amounts. Then also the exhaust gas amounts are larger, which will affect the whole function of the pre-heating system as a whole. If the results show a decline in air demand, then this gives extra capacity on the fan side, giving the opportunity to increase production levels. It is also of great importance to maintain the product quality at all times; therefore, the conditions (oxidising/reducing, temperature) in the kiln system must be carefully controlled.

Thus, if there are doubts about whether one should change fuels in the rotary kiln system, it may be of use to simulate the possible cases beforehand and obtain information on how serious the changes involved might be. This can be also used as to check whether the equipment is suitable and flexible enough for the new fuel combinations.

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References

- [1] O. Marin, O. Charon, J. Dugue, S. Dukhan, W. Zhou, *Combust. Sci. Technol.* 164 (2001) 193–207.
- [2] E. Mastorakos, A. Massias, C.D. Tsakiroglou, D.A. Goussis, V.N. Burganos, A.C. Payatakes, *Appl. Math. Model.* 23 (1999) 55–76.
- [3] D. Giddings, S.J. Pickering, K. Simmons, C.N. Eastwick, *J. Inst. Energy* 75 (2002) 91–99.
- [4] P.A. Alsop, *Cement Plant Operations Handbook for Dry Process Plants*, International Cement Review, Dorking, Surrey, 2nd ed., 1998.
- [5] U. Kääntee, R. Zevenhoven, R. Backman, M. Hupa, *Proceedings of R'2000 Recovery-Recycling-Reintegration*, Toronto, Canada, June (CD-ROM), 2000, p. 1070.
- [6] P.C. Hewlett, *Lea's Chemistry of Cement and Concrete*, Arnold, London, 1998, p. 78.
- [7] U. Kääntee, R. Zevenhoven, R. Backman, M. Hupa, *Proceedings Recycling and Reuse of Used Tyres*, Dundee, Scotland, 2001 March, pp. 81–92.
- [8] S. Enestam, *Advanced Phase Equilibrium Calculations as a Tool for Analysing Ash-Related Fireside Problems in Combustion and Gasification Processes*, Lic. Tech. Thesis, Åbo Akademi University, Turku, Finland, 1999.